

```

1  ESTREL Main Analysis Code
2
3  ## load packages
4  library(ggplot2)
5  library(mice)
6
7  ## load data
8  load("dat/dat.rda") ## raw data
9  load("dat/mice0.rds") ## imputations on FAS
10 N.imp <- length(mice0$imp[[1]]) ## number of imputations used
11
12 ## plot raw data
13 ggplot(data,
14         aes(x = medication, y = my.data[, ep])) +
15         geom_boxplot(outlier.colour = "white") + geom_jitter() +
16         theme_bw() + ylab(ep.name)
17
18 ## build model
19 # fma_affected.Visit_3: primary endpoint
20 # fma_affected.Visit_1.cent: baseline value (centered)
21 # medication: treatment group acc. to randomization
22 model <- with(mice0,
23              lm(fma_affected.Visit_3 ~ fma_affected.Visit_1.cent + medication))
24
25 ## pool and return model
26 summary(pool(model))
27
28 ## do diagnostic plots
29 par(mfrow = c(2,2))
30 for(i in c(1:N.imp)){
31   m <- lm(fma_affected.Visit_3 ~ fma_affected.Visit_1.cent + medication,
32          data = complete(my.data = mice0, i))
33   plot(m, main = paste0("Imputation: ", i))
34 }
35 par(mfrow = c(1,1))

```